

# Cut-Cut Commutations Are Not Completely Superfluous

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## Abstract

In the definition of a cut-elimination procedure, there often is a *cut* – *cut* commutation allowing to swap two *cut*-rules. This allows for instance to fully eliminate a chosen *cut*-rule without reducing other *cut*-rules in a proof. One may wonder whether such *cut* – *cut* steps are superfluous. We prove it is not in linear logic: some cut-free forms of a proof can only be reached using a *cut* – *cut* commutation.

## 1 Introduction

We consider the sequent calculus of linear logic [Gir87]. As many logics, it has a *cut*-rule and a rewriting system called *cut-elimination*, which details how to reach a cut-free (or normal) proof starting from a proof with possibly many *cut*-rules. Cut-elimination has been studied extensively in this system and more generally in linear logic (but mainly for its proof-net syntax): mostly its normalization [Acc13; DG99; LM08; Tor03; PT10] but also its confluence [CP05; Di24]. In particular, it is well-known one can reach a normal form without using any *cut* – *cut* commutations. It can also be proved that equality up to cut-elimination is the same as equality up to cut-elimination without the *cut* – *cut* commutation. One may wonder whether such *cut* – *cut* steps are completely superfluous: can the same normal forms be reached with and without *cut* – *cut* commutations? We answer this question negatively, exhibiting a simple counter-example. This counter-example is even one for the simpler multiplicative-exponential fragment of linear logic, that we will consider here for simplicity's sake.

## 2 Definitions

We consider the sequent calculus of multiplicative-exponential linear logic [Gir87]. Formulas are given by the following grammar, with  $X$  an atom in a given countable set:

$$A, B ::= X^+ \mid X^- \mid A \wp B \mid A \otimes B \mid \perp \mid 1 \mid ?A \mid !A$$

We define on formulas a function  $(\cdot)^\perp$  called **orthogonality**, also named negation or duality, through the following inductive definition:

$$\begin{array}{c}
\frac{}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash A^\perp, \Gamma \quad \vdash A, \Delta}{\vdash \Gamma, \Delta} \text{ (cut)} \\
\frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} \text{ (\wp)} \quad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \text{ (\otimes)} \quad \frac{\vdash \Gamma}{\vdash \perp, \Gamma} \text{ (\perp)} \quad \frac{}{\vdash 1} \text{ (1)} \\
\frac{\vdash A, \Gamma}{\vdash ?A, \Gamma} \text{ (?d)} \quad \frac{\vdash ?A, ?A, \Gamma}{\vdash ?A, \Gamma} \text{ (?c)} \quad \frac{\vdash \Gamma}{\vdash ?A, \Gamma} \text{ (?w)} \quad \frac{\vdash A, ?\Gamma}{\vdash !A, ?\Gamma} \text{ (!)}
\end{array}$$

Figure 1: Rules of Linear Logic

$$\begin{array}{ll}
(X^+)^\perp = X^- & (X^-)^\perp = X^+ \\
(A \wp B)^\perp = B^\perp \otimes A^\perp & (A \otimes B)^\perp = B^\perp \wp A^\perp \\
\perp^\perp = 1 & 1^\perp = \perp \\
(?A)^\perp = !A^\perp & (!A)^\perp = ?A^\perp
\end{array}$$

Sequents are sets of (occurrences of) formulas written as  $\vdash A_1, \dots, A_n$ . Rules of linear logic are given on Fig. 1, where  $A$  and  $B$  stand for arbitrary formulas,  $\Gamma$  and  $\Delta$  for sets of (occurrences of) formulas. By  $?\Gamma$  we mean that each formula of  $\Gamma$  is a  $?$ -formula.

As in many systems with a *cut*-rule, the *cut*-rule is *admissible* in this logic: the same sequents can be proved with and without the *cut*-rule. *Cut-elimination* is the procedure turning a proof into a *cut*-free one. As we will not use *ax*-  $\otimes$ -  $\wp$ -  $?_c$ - and  $?_w$ -rules, we give this rewriting system without those. For a full description, see [Di24, Chapter 1].

**Definition 1. Cut-elimination** is the rewriting system whose rules are described on Table 1, up to commuting the two branches of any *cut*-rule.

The “up to commutation” means we consider a version of each case with the left and right premises of any *cut*-rule swapped – *e.g.* the *cut* – *cut* case gives 4 rewriting rules.

### 3 Cut-cut commutations allow to reach more normal forms

We show that removing the *cut* – *cut* step in multiplicative-exponential linear logic (and thus in linear logic) narrows the set of normal forms.

**Lemma 2.** *There exist a proof  $\pi$  and a cut-free proof  $\phi$  such that  $\pi$  reduces by cut-elimination to  $\phi$  using a cut – cut elimination step (and other cut elimination steps), but  $\pi$  does not reduce to  $\phi$  without using a cut – cut elimination step.*



Without using *cut – cut* steps, cut-elimination on  $\pi$  leads to a unique normal form:

$$\frac{\frac{\frac{}{\vdash 1} (1)}{\vdash ?1} (?_d)}{\vdash ?1, \perp} (\perp)$$

For our purpose, it is enough to prove any normal form of  $\pi$  reached without a *cut – cut* step has a  $\perp$ -rule at its root. This is the case: one first has to apply a  $\perp – cut$  step on the upper *cut*-rule of  $\pi$ ; then, whatever happens on the upper *cut*-rule, the first step involving the bottom *cut*-rule must be a  $\perp – cut$  step, resulting in a  $\perp$ -rule at the root of the proof. Meanwhile, consider the result of applying a *cut – cut* step first:

$$\frac{\frac{\frac{\frac{\frac{}{\vdash 1} (1)}{\vdash 1, \perp} (\perp)}{\vdash 1, ?\perp} (?_d)}{\vdash ?1, ?\perp} (?_d)}{\frac{\frac{\frac{}{\vdash 1} (1)}{\vdash ?1} (?_d) \quad \frac{\frac{\frac{}{\vdash \perp, ?1, ?\perp} (\perp)}{\vdash !\perp, ?1, ?\perp} (!)}{\vdash ?1, ?\perp} (cut)}{\vdash ?1, \perp} (cut) \quad \frac{\frac{\frac{}{\vdash 1} (1)}{\vdash !1} (!)}{\vdash !1, \perp} (\perp)}{\vdash ?1, \perp} (cut)$$

Here, one can apply a  $?_d – !$  step on the upper *cut*-rule, then a  $\perp – 1$  step still on this *cut*-rule, followed by a  $?_d – cut$  step on the (bottom) *cut*-rule, leading to a proof with a  $?_d$ -rule at its root. In particular, one can reach the following normal form  $\phi$ :

$$\frac{\frac{\frac{}{\vdash 1} (1)}{\vdash 1, \perp} (\perp)}{\vdash ?1, \perp} (?_d)$$

□

## 4 Conclusion

Perhaps surprisingly, *cut – cut* commutative steps of cut-elimination are not totally useless: they allow to reach more normal forms in multiplicative-exponential linear logic, and so in linear logic. Nonetheless, our example seems hard to adapt without a contextual rule such as  $!$  – which needs the sequent it is applied on to be of the shape  $!A, ?\Gamma$ . Hence, we conjecture that in the multiplicative-additive fragment of linear logic the same normal forms can be reached with and without *cut – cut* commutative steps.

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